

Securing the Grid: Sequential Anomaly Detection in Industrial Control System Traffic Using LSTM Autoencoders

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Abstract

As power grids become increasingly digitized, cybersecurity vulnerabilities in industrial control systems (ICS) pose serious risks to energy infrastructure. Protocols like Modbus govern communications within SCADA networks, yet they were not designed with security in mind, making them attractive targets for cyberattacks. This study, conducted at Brookhaven National Laboratory, develops a deep learning approach to detect anomalous network behavior in Modbus traffic. Using the CIC Modbus Dataset 2023, network features were extracted from packet capture files via TShark and fed into a Long Short-Term Memory (LSTM) Autoencoder, a neural network architecture well-suited for learning patterns in sequential data. By training exclusively on normal traffic, the model flags deviations through reconstruction error, achieving 99.98% detection accuracy. This work contributes a validated anomaly detection component toward broader AI-driven energy security research, directly supporting the Department of Energy's Genesis Mission to accelerate discovery science and strengthen national security across its 17 national laboratories.

Introduction

The reliable operation of critical infrastructure, including power generation, water treatment, and transportation, depends on the security of industrial control systems (ICS). Historically, ICS networks were physically isolated from external threats. However, the widespread adoption of networked SCADA systems and smart grid technologies has exposed operational technology (OT) networks to malicious actors who previously had no viable entry point.³ High-profile incidents such as the 2015 and 2016 Ukrainian power grid attacks and the 2021 Florida water treatment intrusion demonstrated that ICS cyberattacks can carry severe physical consequences.⁴ These events have driven significant investment in ICS-specific intrusion detection systems. While labeled datasets like the CIC Modbus Dataset 2023 exist,^{1,2} unsupervised methods offer complementary detection for novel or zero-day attacks absent from training labels, motivating the approach used here. This paper presents an experimental study conducted at Brookhaven National Laboratory evaluating an LSTM Autoencoder for anomaly detection on Modbus network traffic. The Modbus protocol, developed in 1979, governs sensor polling and command issuance to programmable logic controllers (PLCs), producing regular request-response patterns well-suited for behavioral modeling.³ Prior deep learning work has established LSTM autoencoders as effective for sequential ICS data,^{5,6,7,8} exploiting the principle that anomalous inputs generate elevated reconstruction error. This study evaluates whether that approach can reliably distinguish normal Modbus communication from attack traffic without labeled training data, contributing a validated unsupervised detection component toward a broader cross-modal foundational model for power grid cybersecurity.

Methods

- A. Dataset**
- Traffic sourced from the **CIC Modbus Dataset 2023**^{1,2}.
 - Simulated substation environment with **benign + 9 attack categories** (e.g., false data injection, brute force write, baseline replay).
 - Three attack configurations: **External Attacker**, **Compromised IED**, **Compromised SCADA HMI**.
 - Docker-based testbed simulating IEDs and SCADA HMIs over Modbus/
- B. Preprocessing**
- Raw PCAPs parsed with **TShark/PyShark** → extracted timestamps, IPs, protocol, payload length, Modbus function codes & register addresses.
 - Packets grouped into **sliding windows of 50 consecutive packets** per protocol stream to preserve temporal ordering.
 - Benign windows** → **training only**; **labeled attack windows** → **evaluation only**.

Feature	Spearman Correlation
bit_count	0.9820
reference_num	-0.9755
word_count	0.9607
function_code	0.9201
requests_per_second	0.8642

Table 1: Top 5 features by absolute Spearman correlation with the anomaly label (External Attack scenario).

- C. Feature Engineering**
- Started with **48 available flow features** from the dataset.
 - Spearman correlation analysis → narrowed to **17 candidate features** (Figure 1).
 - Final model uses **top 5 most predictive features** (Table 1).
 - Features capture both **packet structure** (Modbus fields) and **session volume** (rate behavior).
 - Normalized to **[0, 1]** via Min-Max scaling fit on training data only,

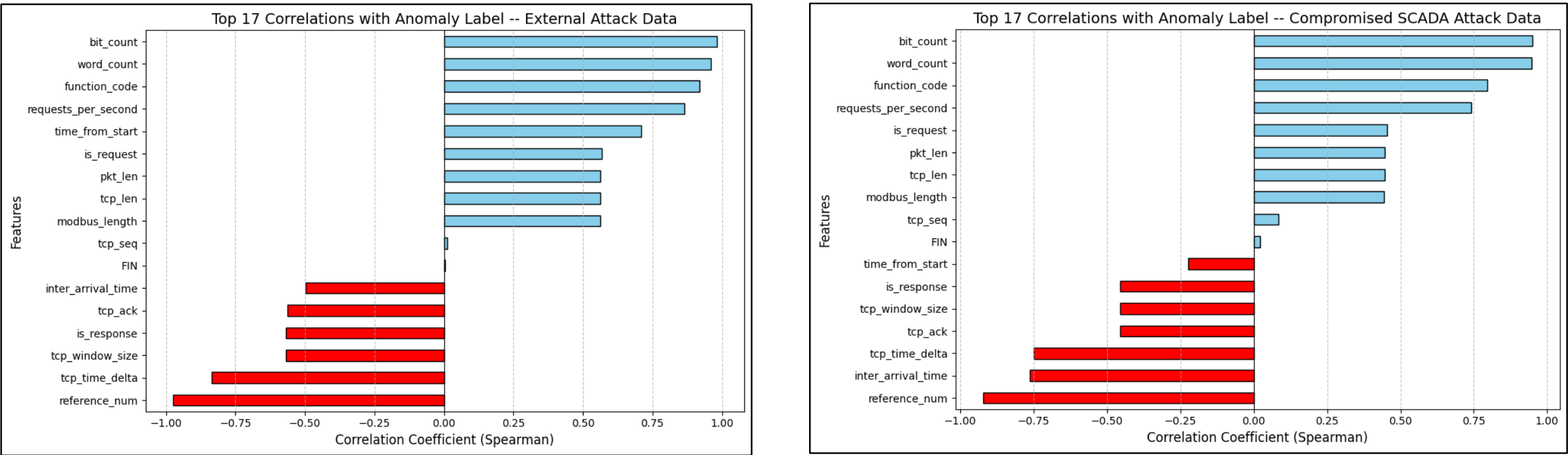


Figure 1: Top 17 Spearman correlation coefficients between extracted features and the anomaly label. (Left) External Attack scenario. (Right) Compromised SCADA scenario. Positive values (blue) indicate features more present in anomalous traffic; negative values (red) indicate features more present in benign traffic.

- D. Model Architecture & Training**
- LSTM Autoencoder** (PyTorch): encoder compresses → decoder reconstructs input sequences.
 - Encoder**: 2-layer LSTM, hidden dim = 64.
 - Decoder**: symmetric 2-layer LSTM reconstructing from repeated final encoder hidden state.
 - Input shape: **(batch × 50 timesteps × 5 features)**.
 - Trained **only on benign traffic** using MSE reconstruction loss.
 - Optimizer**: Adam | LR = 1×10^{-4} | Batch size = 64.
 - Scheduler**: ReduceLROnPlateau (patience = 5, decay = 0.5) | Gradient clipping (max norm = 1.0).
 - 50 epochs**, 80/20 train-validation split on benign data.
 - Anomaly flagging**: per-sequence reconstruction error exceeding the **95th or 99th percentile** of the training error distribution.

Conclusion

This study applied an LSTM Autoencoder for unsupervised anomaly detection on Modbus ICS traffic, trained with no labeled data yet evaluated across nine attack categories. At the 95th percentile threshold, the model achieved AUC-ROC of 1.0000 and F1 of 0.98 on External Attack, and AUC-ROC of 0.9983 and F1 of 0.92 on Compromised SCADA; stricter 99th percentile thresholding pushed F1 to 0.99 and 0.94 respectively. These results confirm sequence-aware deep learning as a viable ICS intrusion detection approach, with complementary value to supervised methods for detecting novel or zero-day attacks. Future work should benchmark supervised baselines on the same dataset and explore hybrid architectures for real-time substation monitoring.

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Results

- LSTM Autoencoder achieved **strong detection on both test sets** (Tables 2 & 3).
- External Attack**: AUROC = **1.0000** (perfect separation).
- Compromised SCADA**: AUROC = **0.9983**.
- Raising threshold from **95th** → **99th percentile** reduced false positives while maintaining near-perfect recall on both sets.
- Compromised SCADA harder to detect**: precision at 95th percentile = **0.80** vs. **0.96** on External Attack, suggesting compromised SCADA traffic patterns more closely resemble normal behavior.
- Figure 2 shows reconstruction error over time for both scenarios; **gray regions = true anomaly windows**.

Scenario	Precision	Recall	F1-Score	Accuracy	AUC-ROC
External Attack	0.96	1.00	0.98	0.98	1.0000
Compromised SCADA	0.80	1.00	0.89	0.92	0.9983

Table 2: Anomaly detection performance using the 95th percentile threshold (0.0144).

Scenario	Precision	Recall	F1-Score	Accuracy	AUC-ROC
External Attack	0.99	1.00	0.99	0.99	1.0000
Compromised SCADA	0.84	1.00	0.91	0.94	0.9983

Table 3: Anomaly detection performance using the 99th percentile threshold (0.0304).

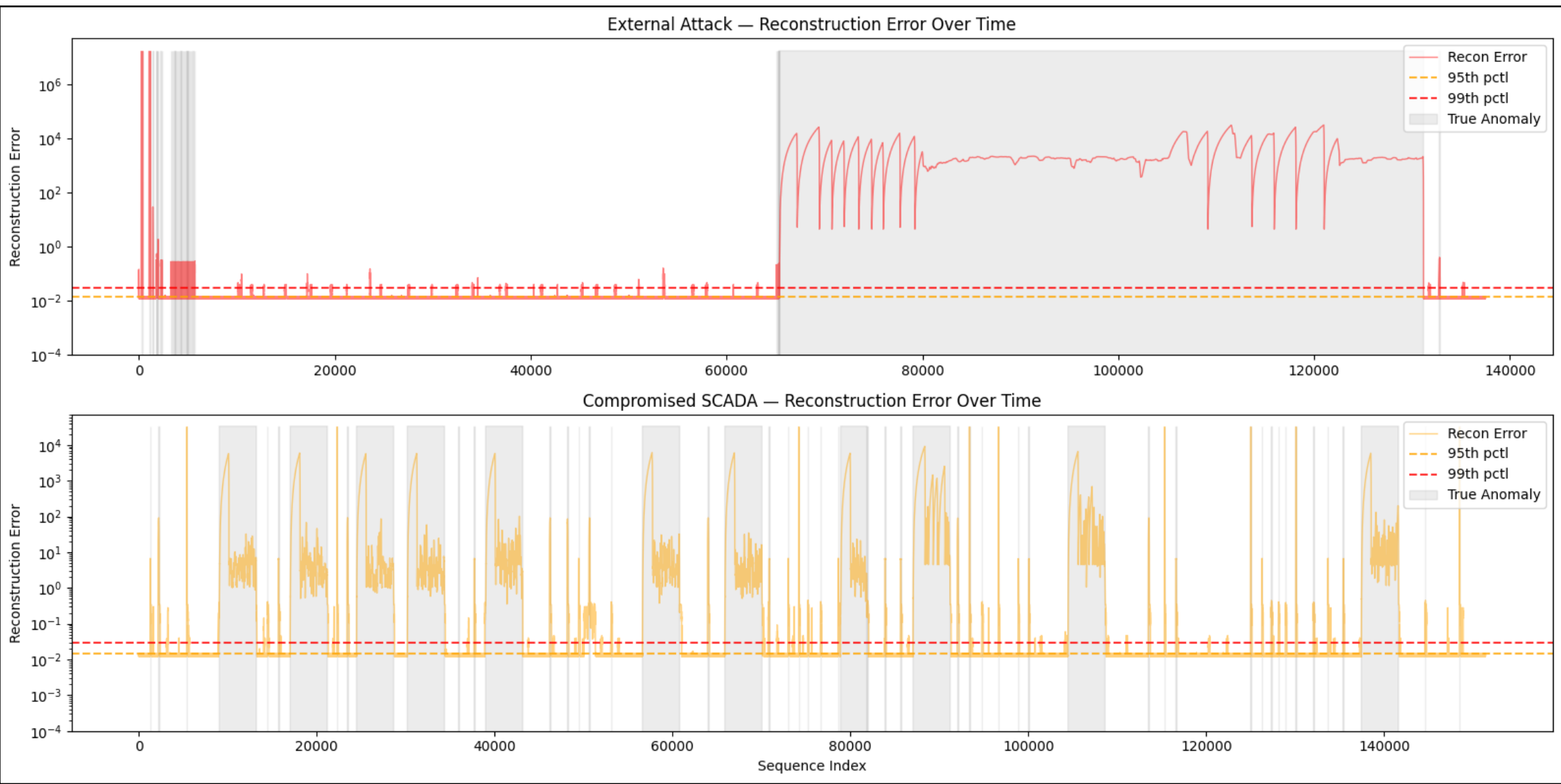


Figure 2: Reconstruction error over time for the (top) External Attack and (bottom) Compromised SCADA test sets. Dashed lines indicate the 95th and 99th percentile thresholds. Gray-shaded regions represent true anomaly periods.

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